

Python Programming Fundamentals

Why Collections?

Programming Fundamentals Team

SETU

2026-10-10



Outline

1. The Problem

2. Python Lists as Arrays

3. Why This Matters

4. Today's Goal



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1. The Problem

So far, we can store one value at a time.

```
student1 = "Ava"
student2 = "Ben"
student3 = "Cara"
```



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```
student1 = "Ava"
student2 = "Ben"
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```

This works for three students, but not for thirty.

Collections let us store many related values under one name.



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2. Python Lists as Arrays

In many languages, students first meet **arrays**.

In Python, the closest beginner-friendly equivalent is a **list**.

```
students = ["Ava", "Ben", "Cara"]
grades = [72, 81, 65, 90]
```



2. Python Lists as Arrays

In many languages, students first meet **arrays**.

In Python, the closest beginner-friendly equivalent is a **list**.

```
students = ["Ava", "Ben", "Cara"]
grades = [72, 81, 65, 90]
```

A list is:

- ordered
- changeable
- indexed from 0
- able to grow and shrink



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3. Why This Matters

Collections are used everywhere:

- grades in a class
- items in a shopping basket
- sensor readings
- names in a booking system
- records loaded from a file



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Collections are used everywhere:

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They also form the basis of later topics:

- files and JSON
- CRUD programs
- classes and objects
- data processing



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By the end of this session, you should be able to:

- create a list
- access items by position
- change list contents
- use strings as sequences
- slice lists and strings
- process lists using loops



Thanks for Watching - Any questions?

